HALF - RANGE SERIES

SINE SERIES: If it is required to expand $f(x)$ as a sine series in $0 < x < c$; then we extend the function reflecting it in the origin, so that $f(-x) = -f(x)$

Then the extended function is odd in $(-c, c)$ and its expansion will give the desired Fourier sine series:

$$f(x) = \sum_{n=1}^{\infty} \left(b_n \sin \frac{n\pi x}{c}\right) \quad \text{where } b_n = \frac{2}{c}\int_0^c f(x).\sin\frac{n\pi x}{c}\,dx$$

COSINE SERIES: If it is required to expand $f(x)$ as a cosine series in $0 < x < c$, then we extend the function reflecting it in the y – axis, so that $f(-x) = f(x)$.

Then the extended function is even in $(-c, c)$ and its expansion will give the desired Fourier cosine series. $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n cos\frac{n\pi x}{c}\right)$ Where $a_0 = \frac{2}{c}\int_0^c f(x)dx$ and $a_n = \frac{2}{c}\int_0^c f(x).\cos\frac{n\pi x}{c}dx$

Note: We can obtain half range sine series, and half range cosine series, both of the same function $f(x)$ in $(0, c)$. In such cases the graphs of $f(x)$in $(0, c)$ are the same but outside $(0, c)$ are different for the sine and cosine series. That is why we get different forms of series for the same function.

1. Obtain half – range cosine series of period 2π for a function $f(x) = \sin x \ in \ [0, \pi]$. Hence deduce that

(i) $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$ **(ii)** $\frac{1}{2} = \frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \cdots$

(iii) $\frac{\pi^2-8}{16} = \frac{1}{1^2.3^2} + \frac{1}{3^2.5^2} + \frac{1}{5^2.7^2} + \cdots$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx$ $[\because l = \pi]$

$$a_0 = \frac{2}{\pi}\int_0^\pi f(x)\,dx = \frac{2}{\pi}\int_0^\pi \sin x\,dx = \frac{2}{\pi}[-\cos x]_0^\pi = \frac{4}{\pi}$$

$$a_n = \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx = \frac{1}{\pi}\int_0^\pi \sin x \cos nx\,dx \quad \text{............. (1)}$$

$$= \frac{1}{\pi}\int_0^\pi [\sin(1+n)\,x + \sin(1-n)\,x]\,dx$$

$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n}\right]_0^\pi$$

$$= \frac{1}{\pi}\left[-\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1}\right]_0^\pi$$

$$= \frac{1}{\pi}\left[-\frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1} + \frac{1}{n+1} - \frac{1}{n-1}\right]$$

$a_n = 0$ if n is odd and $n \neq 1$. And

$a_n = \frac{-2}{\pi}\left[-\frac{1}{n+1} + \frac{1}{n-1}\right]$ if n is even

$= -\frac{4}{\pi}\left(\frac{1}{n^2-1}\right)$ if n is even

∴ If $n = 1$ from (1), we get,

$$a_1 = \frac{2}{\pi}\int_0^\pi \sin x \cos x\,dx = \frac{1}{\pi}\int_0^\pi \sin 2x\,dx = \frac{1}{\pi}\left[-\frac{\cos 2x}{2}\right]_0^\pi = 0$$

$$\therefore f(x) = \sin x = \frac{2}{\pi} - \frac{4}{\pi}\left(\frac{\cos 2x}{3} + \frac{\cos 4x}{15} + \frac{\cos 6x}{35} + \cdots\right) \quad \text{............. (2)}$$

(i) The series (2) can be written as $\sin x = \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{(2n)^2-1}\cos 2nx$

Now, putting $x=\frac{\pi}{2}$, we get

$$1=\frac{2}{\pi}-\frac{4}{\pi}\sum_{n=1}^{\infty}\frac{1}{2}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\cos n\pi$$

$$\frac{\pi}{2}=1-2\sum_{n=1}^{\infty}\frac{1}{2}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)(-1)^n$$

$$=1-2\left[\frac{1}{2}\left\{-\left(\frac{1}{1}-\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)-\left(\frac{1}{5}-\frac{1}{7}\right)+\left(\frac{1}{7}-\frac{1}{9}\right)-\cdots\right\}\right]$$

$$=1-\left[-\left(\frac{1}{1}+\frac{1}{3}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)-\left(\frac{1}{5}-\frac{1}{7}\right)+\left(\frac{1}{7}-\frac{1}{9}\right)+\cdots\right]$$

$$=1+\left[1-\frac{1}{3}-\frac{1}{3}+\frac{1}{5}+\frac{1}{5}-\frac{1}{7}-\frac{1}{7}+\cdots\right]=2\cdot\left[1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots\right]$$

$$\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots$$

(ii) Putting $x=0$ in (2) we get,

$$-\frac{2}{\pi}=-\frac{4}{\pi}\left[\frac{1}{3}+\frac{1}{15}+\frac{1}{35}+\cdots\right]$$

$$\therefore\frac{1}{2}=\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\cdots$$

(iii) By Paresval identity, we have

$$\frac{2}{\pi}\int_0^{\pi}[f(x)]^2\,dx=\frac{{a_0}^2}{2}+\left[{a_1}^2+{a_2}^2+{a_3}^2+\cdots\right]$$

Now, $\frac{2}{\pi}\int_0^{\pi}[f(x)]^2\,dx=\frac{2}{\pi}\int_0^{\pi}\sin^2 x\,dx=\frac{2}{\pi}\int_0^{\pi}\left(\frac{1-\cos 2x}{2}\right)dx=\frac{1}{\pi}\left[x-\frac{\sin 2x}{2}\right]_0^{\pi}=\frac{1}{\pi}[\pi]=1$

$$\therefore 1=\frac{1}{2}\cdot\frac{16}{\pi^2}+\left[\frac{16}{\pi^2}\left\{\frac{1}{3^2}+\frac{1}{15^2}+\frac{1}{35^2}+\cdots\right\}\right]$$

$$1=\frac{8}{\pi^2}+\frac{16}{\pi^2}\left[\frac{1}{1^2\cdot3^2}+\frac{1}{3^2\cdot2^2}+\frac{1}{5^2\cdot7^2}+\cdots\right]$$

$$1-\frac{8}{\pi^2}=\frac{16}{\pi^2}\left[\frac{1}{1^2\cdot3^2}+\frac{1}{3^2\cdot2^2}+\frac{1}{5^2\cdot7^2}+\cdots\right]$$

$$\frac{\pi^2-8}{16}=\frac{1}{1^2\cdot3^2}+\frac{1}{3^2\cdot2^2}+\frac{1}{5^2\cdot7^2}+\cdots$$

2. Express $f(x)$as half rage sine series in $(0,\pi)$, Where $f(x)=\begin{cases}x, & 0<x<\pi/2\\ \pi-x & \frac{\pi}{2}<x<\pi\end{cases}$

Hence deduce that **(i)** $\frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+\cdots=\frac{\pi^4}{96}$ **(ii)** $\frac{\pi^2}{8}=\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\cdots$

Solution: Let $f(x)=\sum b_n\sin nx$ $\quad[\because l=\pi]$

$$\therefore b_n=\frac{2}{\pi}\left[\int_0^{\pi/2}x\sin nx\,dx+\int_{\pi/2}^{\pi}(\pi-x)\sin nx\,dx\right]$$

$$=\frac{2}{\pi}\left[\left\{x\left(-\frac{\cos nx}{n}\right)-\left(-\frac{\sin nx}{n^2}\right)(1)\right\}_0^{\pi/2}+\left\{(\pi-x)\left(-\frac{\cos nx}{n}\right)-\left(-\frac{\sin nx}{n^2}\right)(-1)\right\}_{\pi/2}^{\pi}\right]$$

$$=\frac{2}{\pi}\left[\left\{-\frac{\pi}{2}\frac{\cos(n\pi/2)}{n}+\frac{\sin(n\pi/2)}{n^2}-0-0\right\}+\left\{0-0+\frac{\pi}{2}\frac{\cos(n\pi/2)}{n}+\frac{\sin(n\pi/2)}{n^2}\right\}\right]$$

$$=\frac{4}{\pi}\frac{\sin(n\pi/2)}{n^2}$$

$$\therefore b_1=\frac{4}{\pi},b_2=0,b_3=-\frac{4}{\pi}\cdot\frac{1}{3^2},b_4=0\ldots\ldots\ldots$$

$$\therefore f(x)=\frac{4}{\pi}\left[\frac{1}{1^2}\sin x-\frac{1}{3^2}\sin 3x+\frac{1}{5^2}\sin 5x-\frac{1}{7^2}\sin 7x+\cdots\right]$$

(i) By Parseval's identity $\frac{2}{\pi}\int_0^{\pi}[f(x)]^2\,dx=\left[{b_1}^2+{b_2}^2+{b_3}^2+\cdots\infty\right]$

$$\therefore \frac{2}{\pi}\int_0^{\pi}[f(x)]^2\,dx = \frac{2}{\pi}\left[\int_0^{\pi/2} x^2\,dx + \int_{\pi/2}^{\pi}(\pi-x)^2\,dx\right]$$
$$= \frac{2}{\pi}\left[\int_0^{\pi/2} x^2\,dx + \int_{\pi/2}^{\pi}(\pi^2-2\pi x+x^2)dx\right]$$
$$= \frac{2}{\pi}\left[\left\{\frac{x^3}{3}\right\}_0^{\pi/2} + \left\{\pi^2 x - \pi x^2 + \frac{x^3}{3}\right\}_{\pi/2}^{\pi}\right]$$
$$= \frac{2}{\pi}\left[\left\{\frac{\pi^3}{24}-0\right\} + \left\{\left(\pi^3-\pi^3+\frac{\pi^3}{3}\right) - \left(\frac{\pi^3}{2}-\frac{\pi^3}{4}+\frac{\pi^3}{24}\right)\right\}\right]$$
$$= \frac{2}{\pi}\cdot\frac{\pi^3}{12} = \frac{\pi^2}{6}$$
$$\therefore \frac{\pi^2}{6} = \left[\frac{16}{\pi^2}\cdot\frac{1}{1^4} + \frac{16}{\pi^2}\cdot\frac{1}{3^4} + \frac{16}{\pi^2}\cdot\frac{1}{5^4} + \cdots\right]$$
$$\therefore \frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots$$

(ii) For deduction put $x = \pi/2$,

$$\therefore \frac{\pi}{2} = \frac{4}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right]$$
$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$$

3. Obtain half – range Fourier cosine series for $f(x) = x$ in $0 < x < 2$. Hence deduce that

(i) $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots = \frac{\pi^4}{96}$ **(ii)** $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots = \frac{\pi^4}{90}$

Or Show that $(0, l)$ $x = \frac{l}{2} - \frac{4l}{\pi^2}\left[\frac{1}{1^2}cos\frac{\pi x}{l} + \frac{1}{3^2}cos\frac{3\pi x}{l} + \frac{1}{5^2}cos\frac{5\pi x}{l} + \cdots.\right]$

Hence deduce that **(i)** $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots = \frac{\pi^4}{90}$ **(ii)** $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \cdots = \frac{\pi^4}{96}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi x}{l}\right)$. Here, $l = 2$

$$\therefore a_0 = \frac{2}{l}\int_0^l f(x)\,dx = \int_0^2 x\,dx = \left[\frac{x^2}{2}\right]_0^2 = 2$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx = \frac{2}{2}\int_0^2 x\cos\frac{n\pi x}{2}\,dx$$
$$= \left[x\frac{\sin(n\pi x/2)}{n\pi/2} + \frac{\cos(n\pi x/2)}{n^2\pi^2/2^2}\cdot 1\right]_0^2$$
$$= \left[2\cdot(0) + \frac{\cos n\pi}{n^2\pi^2/2^2} - 0 - \frac{1}{n^2\pi^2/2^2}\right] = \frac{[(-1)^n-1]}{n^2\pi^2/2^2}$$
$$= \begin{cases} -4\cdot\frac{2}{n^2\pi^2} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even}\end{cases}$$

$$\therefore x = 1 - \frac{8}{\pi^2}\left[\frac{1}{1^2}\cos\frac{\pi x}{2} + \frac{1}{3^2}\cos\frac{3\pi x}{2} + \cdots\right]$$

(i) By Parseval's identity $\frac{2}{l}\int_0^l [f(x)]^2\,dx = \frac{{a_0}^2}{2} + [{a_1}^2 + {a_2}^2 + \cdots]$

$$\therefore \text{LHS} = \frac{2}{2}\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}$$
$$\therefore \frac{8}{3} = \frac{4}{2} + \frac{64}{\pi^4}\left\{\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\right\}$$
$$\frac{8}{3} - 2 = \frac{64}{\pi^4}\left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\right]$$

$$\frac{\pi^4}{96} = \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\right]$$

(ii) Let $S = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots$

$$= \left(\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots\right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \cdots\right)$$

$$= \left(\frac{\pi^4}{96}\right) + \frac{1}{2^4}\left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots\right)$$

$\therefore S = \frac{\pi^4}{96} + \frac{S}{16}$ $\qquad \therefore S = \frac{\pi^4}{90}$

4. Find half – range sine series for $f(x) = x \sin x$ in $(0, \pi)$ and deduce that

(i) $\frac{1}{1^2.3^2} - \frac{3}{5^2.7^2} + \frac{5}{9^2.11^2} - \frac{7}{13^2.15^2} + \cdots = \frac{\pi^2}{64\sqrt{2}}$

(ii) $\frac{1}{1^2} - \frac{1}{3^2} - \frac{1}{5^2} + \frac{1}{7^2} + \frac{1}{9^2} - \frac{1}{11^2} + \cdots = \frac{\pi^2}{8\sqrt{2}}$

Solution: Let $f(x) = \sum b_n \sin nx$ $\qquad [\because l = \pi]$

$\therefore b_n = \frac{2}{\pi}\int_0^\pi x \sin x \sin nx \, dx$ $\qquad$ .... (1)

$$= -\frac{1}{\pi}\int_0^\pi x[\cos(n+1)\,x - \cos(n-1)\,x]\,dx$$

$$= -\frac{1}{\pi}\left[x\left\{\frac{\sin(n+1)x}{n+1} - \frac{\sin(n-1)x}{n-1}\right\} - \{1\}\left\{-\frac{\cos(n+1)x}{(n+1)^2} + \frac{\cos(n-1)x}{(n-1)^2}\right\}\right]_0^\pi$$

$$= -\frac{1}{\pi}\left[\pi\left\{\frac{\sin(n+1)\pi}{n+1} - \frac{\sin(n-1)\pi}{n-1}\right\} - (1)\left\{-\frac{\cos(n+1)\pi}{(n+1)^2} + \frac{\cos(n-1)\pi}{(n-1)^2}\right\} - 0 - \frac{1}{(n+1)^2} + \frac{1}{(n-1)^2}\right] \text{ if } n \neq 1$$

Now, $\sin(n \pm 1)\pi = 0$ and $\cos(n \pm 1)\pi = -\cos n\pi$

$$= \frac{1}{\pi}\left[\frac{\cos n\pi}{(n+1)^2} - \frac{\cos n\pi}{(n-1)^2} + \frac{1}{(n+1)^2} - \frac{1}{(n-1)^2}\right]$$

$$= \frac{1}{\pi}\left[\frac{(-1)^n+1}{(n+1)^2} - \frac{(-1)^n+1}{(n-1)^2}\right]$$

$$= \begin{cases} 0 & \text{when } n \text{ is odd and } n \neq 1 \\ -\frac{2}{\pi}\left\{\frac{1}{(n-1)^2} - \frac{1}{(n+1)^2}\right\} & \text{when } n \text{ is even} \end{cases}$$

When $n = 1$ from (1),

$$b_1 = \frac{2}{\pi}\int_0^\pi x \sin^2 x \, dx = \frac{2}{\pi}\int_0^\pi x\left(\frac{1-\cos 2x}{2}\right)dx$$

$$= \frac{1}{\pi}\left[x\left(x - \frac{\sin 2x}{2}\right) - (1)\left(\frac{x^2}{2} + \frac{\cos 2x}{4}\right)\right]_0^\pi$$

$$= \frac{1}{\pi}\left[\pi^2 - 0 - \frac{\pi^2}{2} - \frac{1}{4} + \frac{1}{4}\right] = \frac{\pi}{2}$$

$$\therefore f(x) = x \sin x = \frac{\pi}{2}\sin x - \frac{2}{\pi}\left[\left(\frac{1}{1^2} - \frac{1}{3^2}\right)\sin 2x + \left(\frac{1}{3^2} - \frac{1}{5^2}\right)\sin 4x + \left(\frac{1}{5^2} - \frac{1}{7^2}\right)\sin 6x + \cdots\right]$$

$$\therefore f(x) = \frac{\pi}{2}\sin x + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right)\sin 2x + \left(\frac{1}{5^2} - \frac{1}{3^2}\right)\sin 4x + \left(\frac{1}{7^2} - \frac{1}{5^2}\right)\sin 6x + \cdots\right]$$

(i) For deduction put $x = \frac{\pi}{4}$

$$\therefore \frac{\pi}{4}\sin\frac{\pi}{4} = \frac{\pi}{2}\frac{1}{\sqrt{2}} + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right)\cdot 1 + \left(\frac{1}{5^2} - \frac{1}{3^2}\right)\cdot 0 + \left(\frac{1}{7^2} - \frac{1}{5^2}\right)\cdot(-1) + \cdots\right]$$

$$\frac{\pi}{4}\cdot\frac{1}{\sqrt{2}} = \frac{\pi}{2}\cdot\frac{1}{\sqrt{2}} + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right) + \left(\frac{1}{7^2} - \frac{1}{5^2}\right) + \cdots\right]$$

$$-\frac{\pi}{4\sqrt{2}} = \frac{2}{\pi}\left[\frac{-8}{1^2.3^2} - \frac{-24}{5^2.7^2} + \frac{-40}{9^2.11^2} + \cdots\right]$$

$$-\frac{\pi}{4\sqrt{2}} = \frac{-16}{\pi}\left[\frac{1}{1^2.3^2} - \frac{3}{5^2.7^2} + \frac{5}{9^2.11^2} + \cdots\right]$$
$$\frac{\pi^2}{64\sqrt{2}} = \frac{1}{1^2.3^2} - \frac{3}{5^2.7^2} + \frac{5}{9^2.11^2} - \ \ldots\ldots\ldots$$

(ii) For deduction put $x = \frac{\pi}{4}$

$$\therefore \frac{\pi}{4}\sin\frac{\pi}{4} = \frac{\pi}{2}\frac{1}{\sqrt{2}} + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right)\cdot 1 + \left(\frac{1}{5^2} - \frac{1}{3^2}\right)\cdot 0 + \left(\frac{1}{7^2} - \frac{1}{5^2}\right)\cdot(-1) + \cdots\right]$$
$$\frac{\pi}{4}\cdot\frac{1}{\sqrt{2}} = \frac{\pi}{2}\cdot\frac{1}{\sqrt{2}} + \frac{2}{\pi}\left[\left(\frac{1}{3^2} - \frac{1}{1^2}\right) + \left(\frac{1}{7^2} - \frac{1}{5^2}\right) + \cdots\right]$$
$$-\frac{\pi}{4\sqrt{2}} = \frac{2}{\pi}\left[\frac{1}{3^2} - \frac{1}{1^2} + \frac{1}{7^2} - \frac{1}{5^2} + \cdots\right]$$
$$\frac{\pi^2}{8\sqrt{2}} = \frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots$$

5. In $(0, \pi)$ show that

$$x^2 = \frac{2}{\pi}\left\{\left(\frac{\pi^2}{1} - \frac{4}{1^3}\right)\sin x - \frac{\pi^2}{2}\sin 2x + \left(\frac{\pi^2}{3} - \frac{4}{3^3}\right)\sin 3x - \frac{\pi^2}{4}\sin 4x + \left(\frac{\pi^2}{5} - \frac{4}{5^3}\right)\sin 5x - \cdots\right\}$$

Solution: Let $f(x) = \sum b_n \sin nx$ $\qquad [\because l = \pi]$

$$\therefore b_n = \frac{2}{\pi}\int_0^{\pi} x^2 \sin nx\, dx \qquad \ldots\ldots\ldots\ldots (1)$$
$$= \frac{2}{\pi}\left[(x^2)\left\{-\frac{\cos nx}{n}\right\} - (2x)\left\{-\frac{\sin nx}{n^2}\right\} + (2)\left\{\frac{\cos nx}{n^3}\right\}\right]_0^{\pi}$$
$$= \frac{2}{\pi}\left[\frac{\pi^2}{n}(-\cos n\pi) - 0 + \frac{2}{n^3}(\cos n\pi) - \frac{2}{n^3}\right]$$
$$= \begin{cases} \frac{2}{\pi}\left[\frac{\pi^2}{n} - \frac{4}{n^3}\right] & \text{if } n \text{ is odd} \\ \frac{2}{\pi}\left[-\frac{\pi^2}{n}\right] & \text{if } n \text{ is even} \end{cases}$$
$$\therefore x^2 = \frac{2}{\pi}\left[\left(\frac{\pi^2}{1} - \frac{4}{1^3}\right)\sin x - \frac{\pi^2}{2}\sin 2x + \left(\frac{\pi^2}{3} - \frac{4}{3^3}\right)\sin 3x + \ \ldots\ldots\right]$$

6. Obtain a half range cosine series for $f(x) = \begin{cases} kx, & for\ 0 \le x \le \frac{l}{2} \\ k(l - x) & for\ \frac{l}{2} \le x \le l \end{cases}$

Hence deduce the sum $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi x}{l}\right)$

$$a_0 = \frac{2}{l}\int_0^l f(x)\,dx = \frac{2}{l}\left[\int_0^{l/2} kx\,dx + \int_{l/2}^l k(l - x)\,dx\right] = \frac{2}{l}\left\{\left[k\frac{x^2}{2}\right]_0^{l/2} + \left[klx - k\frac{x^2}{2}\right]_{l/2}^l\right\} = \frac{kl}{2}$$

$$a_n = \frac{2}{l}\left[\int_0^{l/2} kx\cos\frac{n\pi x}{l}dx + \int_{l/2}^l k(l - x)\cos\frac{n\pi x}{l}dx\right]$$
$$= \frac{2}{l}\left[\left\{kx\left(\frac{l}{n\pi}\sin\frac{n\pi x}{l}\right) - k\left(-\frac{l^2}{n^2\pi^2}\cos\frac{n\pi x}{l}\right)\right\}_0^{l/2}\right.$$
$$\left. + \left\{k(l - x)\left(\frac{l}{n\pi}\sin\frac{n\pi x}{l}\right) - (-k)\left(\frac{-l^2}{n^2\pi^2}\cos\frac{n\pi x}{l}\right)\right\}_{l/2}^l\right]$$
$$= \frac{2}{l}\left[\frac{kl^2}{n^2\pi^2}\cos\frac{n\pi}{2} - \frac{kl^2}{n^2\pi^2} - \frac{kl^2}{n^2\pi^2}\cos n\pi + \frac{kl^2}{n^2\pi^2}\cos\frac{n\pi}{2}\right]$$
$$a_1 = 0, a_2 = -\frac{8kl}{2^2\pi^2}, a_3 = 0, a_4 = 0, a_5 = 0, a_6 = -\frac{8kl}{6^2\pi^2}, \ldots\ldots a_{10} = -\frac{8kl}{10^2\pi^2}$$

$\therefore f(x) = \frac{kl}{4} - \frac{8kl}{\pi^2}\left[\frac{1}{2^2}\cos\frac{2\pi x}{l} + \frac{1}{6^2}\cos\frac{6\pi x}{l} + \frac{1}{10^2}\cos\frac{10\pi x}{l} + \cdots\right]$

Now, put $x = l/2$

$\therefore \frac{kl}{2} = \frac{kl}{4} - \frac{8kl}{\pi^2}\left[\frac{1}{2^2}\cos\pi + \frac{1}{6^2}\cos 3\pi + \frac{1}{10^2}\cos 5\pi + \cdots.\right]$

$\therefore \frac{kl}{2} - \frac{kl}{4} = \frac{8kl}{\pi^2}\left[\frac{1}{2^2} + \frac{1}{6^2} + \frac{1}{10^2} + \cdots.\right]$

$\therefore \frac{kl}{4} = \frac{8kl}{\pi^2}\cdot\frac{1}{2^2}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots.\right]$

$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ \ldots\ldots$

7. Obtain the half – range cosine series for $f(x) = \pi x - x^2$ in $(0, \pi)$ and hence deduce that

(i) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots = \frac{\pi^2}{6}$ **(ii)** $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots$

(iii) $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + .. = \frac{\pi^4}{90}$ **(iv)** $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \cdots = \frac{\pi^4}{96}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx$ $\quad [\because l = \pi]$

$\therefore a_0 = \frac{2}{\pi}\int_0^\pi f(x)\,dx = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\,dx = \frac{2}{\pi}\left[\frac{\pi x^2}{2} - \frac{x^3}{3}\right]_0^\pi = \frac{\pi^2}{3}$

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx = \frac{2}{\pi}\int_0^\pi (\pi x - x^2)\cos nx\,dx \\ &= \frac{2}{\pi}\left[(\pi x - x^2)\frac{\sin nx}{n} - (\pi - 2x)\cdot\left(-\frac{\cos nx}{n^2}\right) + (-2)\left(-\frac{\sin nx}{n^3}\right)\right]_0^\pi \\ &= -2\left(\frac{1+\cos n\pi}{n^2}\right) = \begin{cases} 0 & \text{when } n \text{ is odd} \\ -4/n^2 & \text{when } n \text{ is even} \end{cases} \end{aligned}$$

$\therefore x(\pi - x) = \frac{\pi^2}{6} - 4\left[\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + \cdots\right]$

$\therefore x(\pi - x) = \frac{\pi^2}{6} - \left[\frac{1}{1^2}\cos 2x + \frac{1}{2^2}\cos 4x + \frac{1}{3^2}\cos 6x + \cdots\right]$

(i) Now, put $x = 0$

$\therefore 0 = \frac{\pi^2}{6} - \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots.\right)$

$\therefore \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots$

(ii) Again put $x = \frac{\pi}{2}$

$\therefore \frac{\pi^2}{4} = \frac{\pi^2}{6} - \left[-\frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \cdots.\right]$

$\therefore \frac{\pi^2}{4} - \frac{\pi^2}{6} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots$

$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \cdots$

(iii) Now, by Parseval's identity

$\frac{2}{\pi}\int_0^\pi [f(x)]^2\,dx = \frac{{a_0}^2}{2} + [{a_1}^2 + {a_2}^2 + {a_3}^2 + \cdots]$

$\therefore LHS = \frac{2}{\pi}\int_0^\pi [f(x)]^2\,dx = \frac{2}{\pi}\int_0^\pi x^2(\pi - x)^2\,dx$

$= \frac{2}{\pi}\int_0^\pi (\pi^2 x^2 - 2\pi x^3 + x^4)\,dx$

$$= \frac{2}{\pi}\left[\pi^2\frac{x^3}{3} - 2\pi\frac{x^4}{4} + \frac{x^5}{5}\right]_0^\pi$$
$$= \frac{2}{\pi}\left[\frac{\pi^5}{3} - \frac{\pi^5}{2} + \frac{\pi^5}{5}\right]$$
$$= \frac{2}{\pi}\cdot\frac{\pi^5}{30} = \frac{\pi^4}{15}$$
$$\therefore \frac{\pi^4}{15} = \frac{\pi^4}{18} + \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots\right]$$
$$\frac{\pi^4}{15} - \frac{\pi^4}{18} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots$$
$$\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots$$

(iv) $\frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \cdots$
$$\frac{\pi^4}{90} = \left(\frac{1}{1^4} + \frac{1}{3^4} + \ \dots\dots\dots\right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \ \dots\dots\dots\right)$$
$$\frac{\pi^4}{90} = \left(\frac{1}{1^4} + \frac{1}{3^4} + \ \dots\dots\dots\right) + \frac{1}{2^4}\cdot\left(\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \ \dots\dots\dots\right)$$
$$\frac{\pi^4}{90} = \left(\frac{1}{1^4} + \frac{1}{3^4} + \ \dots\dots\dots\right) + \frac{1}{16}\cdot\frac{\pi^4}{90}$$
$$\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \frac{1}{7^4} + \cdots$$

8. Show that, in 0 < x < π, $\cos x = \frac{8}{\pi}\sum_{m=1}^{\infty}\left(\frac{m}{4m^2-1}\right)\sin 2mx$

Solution: We have to obtain half-range sine series for $\cos x$

Let $\cos x = \sum b_n \sin nx$ $\quad [\because l = \pi]$

$$\therefore b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx = \frac{2}{\pi}\int_0^\pi \cos x \sin nx\,dx \qquad \dots\dots\dots\dots (1)$$
$$= \frac{1}{\pi}\int_0^\pi [\sin(1+n)\,x - \sin(1-n)\,x]\,dx$$
$$= \frac{1}{\pi}\left[-\frac{\cos(1+n)x}{1+n} + \frac{\cos(1-n)x}{1-n}\right]_0^\pi$$
$$= \frac{1}{\pi}\left[-\frac{\cos(n+1)\pi}{n+1} - \frac{\cos(n-1)\pi}{n-1} + \frac{1}{n+1} + \frac{1}{n-1}\right]$$
$$= \frac{1}{\pi}\left[\frac{\cos n\pi}{n+1} + \frac{\cos n\pi}{n-1} + \frac{1}{n+1} + \frac{1}{n-1}\right]$$
$$= \frac{1}{\pi}\left[\frac{1}{n+1} + \frac{1}{n-1}\right](1 + \cos n\pi) = \frac{1}{\pi}\cdot\frac{2n}{n^2-1}[1 + (-1)^n]$$
$$= \begin{cases} 0, & \text{if } n \text{ is odd and } n \neq 1 \\ \frac{1}{\pi}\cdot\frac{4n}{n^2-1}, & \text{if } n \text{ is even} \end{cases}$$

When $n = 1$, from (1), we get,

$$b_1 = \frac{2}{\pi}\int_0^\pi \cos x \sin x\,dx = \frac{1}{\pi}\int_0^\pi \sin 2x\,dx = \frac{1}{\pi}\left[-\frac{\cos 2x}{2}\right]_0^\pi = -\frac{1}{2\pi}[1-1] = 0$$

$$\therefore \cos x = \frac{4}{\pi}\left[\frac{2}{2^2-1}\sin 2x + \frac{4}{4^2-1}\sin 4x + \frac{6}{6^2-1}\sin 6x + \cdots.\right]$$
$$= \frac{8}{\pi}\left[\frac{1}{2^2-1}\sin 2x + \frac{2}{4^2-1}\sin 4x + \frac{3}{6^2-1}\sin 6x + \cdots.\right]$$
$$= \frac{8}{\pi}\sum\frac{m}{4m^2-1}\sin m\pi$$

9. Obtain the half – range **(i)** Cosine Series, **(ii)** sine series for $f(x) = lx - x^2$ in $(0, l)$. Hence deduce that

(i) $\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots = \frac{\pi^3}{32}$ **(ii)** $\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \frac{1}{7^6} + .. = \frac{\pi^6}{960}$

(iii) $\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \frac{1}{4^6} + \cdots = \frac{\pi^6}{945}$

Solution: **(i)** **Cosine Series**

Let $f(x) = \frac{a_0}{2} + \sum a_n \cos\frac{n\pi x}{l}$

$$a_0 = \frac{2}{l}\int_0^l (lx - x^2)\,dx = \frac{2}{l}\left[l\frac{x^2}{2} - \frac{x^3}{3}\right]_0^l = \frac{2}{l}\cdot\frac{l^3}{6} = \frac{l^2}{3}$$

$$a_n = \frac{2}{l}\int_0^l (lx - x^2)\cos\frac{n\pi x}{l}\,dx$$

$$= \frac{2}{l}\left[(lx - x^2)\left(\frac{l}{n\pi}\sin\frac{n\pi x}{l}\right) - (l - 2x)\left(-\frac{l^2}{n^2\pi^2}\cos\frac{n\pi x}{l}\right) + (-2)\left(-\frac{l^3}{n^3\pi^3}\sin\frac{n\pi x}{l}\right)\right]_0^l$$

$$= \frac{2}{l}\left[\left\{0 - l\cdot\frac{l^2}{n^2\pi^2}\cos n\pi - 0\right\} - \left\{0 + l\cdot\frac{l^2}{n^2\pi^2} + 0\right\}\right]$$

$$= -\frac{2}{l}\cdot\frac{l^3}{n^2\pi^2}[\cos n\pi + 1]$$

$$= \begin{cases} 0 & \text{if } n \text{ is odd} \\ -\frac{4l^2}{n^2\pi^2} & \text{if } n \text{ is even} \end{cases}$$

$$\therefore f(x) = lx - x^2 = \frac{l^2}{6} - \frac{4l^2}{\pi^2}\left[\frac{1}{2^2}\cos\frac{2\pi x}{l} + \frac{1}{4^2}\cos\frac{4\pi x}{l} + \frac{1}{6^2}\cos\frac{6\pi x}{l} + \cdots\right] \quad \text{..........(1)}$$

(ii) **Sine series**

Let $f(x) = \sum b_n \sin\frac{n\pi x}{l}$

$$b_n = \frac{2}{l}\int_0^l (lx - x^2)\sin\frac{n\pi x}{l}\,dx$$

$$= \frac{2}{l}\left[(lx - x^2)\left(-\frac{l}{n\pi}\cos\frac{n\pi x}{l}\right) - (l - 2x)\left(-\frac{l^2}{n^2\pi^2}\sin\frac{n\pi x}{l}\right) + (-2)\left(\frac{l^3}{n^3\pi^3}\cos\frac{n\pi x}{l}\right)\right]_0^l$$

$$= \frac{2}{l}\left[\left\{0 - 0 - \frac{2l^3}{n^3\pi^3}\cos n\pi - 0\right\} - \left\{0 - 0 - \frac{2l^3}{n^3\pi^3}\right\}\right]$$

$$= \frac{2}{l}\cdot\frac{2l^3}{n^3\pi^3}(-\cos n\pi + 1)$$

$$= \begin{cases} \frac{8l^2}{n^3\pi^3} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases}$$

$$\therefore f(x) = lx - x^2 = \frac{8l^2}{\pi^3}\left[\frac{1}{1^3}\sin\frac{\pi x}{l} + \frac{1}{3^3}\sin\frac{3\pi x}{l} + \frac{1}{5^3}\sin\frac{5\pi x}{l} + \cdots\right] \quad \text{..........(2)}$$

(i) For deduction we put $x = \frac{l}{2}$ in (2)

$$\therefore l\cdot\left(\frac{l}{2}\right) - \left(\frac{l}{2}\right)^2 = \frac{8l^2}{\pi^3}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \cdots\right]$$

$$\therefore \frac{\pi^3}{32} = \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots$$

(ii) Now, by Parseval's identity $\frac{2}{l}\int_0^l [f(x)]^2\,dx = \left[{b_0}^2 + {b_2}^2 + {b_3}^2 + \cdots\infty\right]$

$$\frac{2}{l}\int_0^l [lx - x^2]^2\,dx = \left(\frac{8l^2}{\pi^3}\right)^2\left[\left(\frac{1}{1^3}\right)^2 + \left(\frac{1}{3^3}\right)^2 + \left(\frac{1}{5^3}\right)^2 + \cdots\right]$$

$$\frac{1}{l}\int_0^l (l^2x^2 - 2lx^3 + x^4)\,dx = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6}+\frac{1}{3^6}+\frac{1}{5^6}+\cdots\right]$$

$$\frac{1}{l}\left[l^2\frac{x^3}{3} - 2l\frac{x^4}{4} + \frac{x^5}{5}\right]_0^l = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6}+\frac{1}{3^6}+\frac{1}{5^6}+\cdots\right]$$

$$\therefore \frac{1}{30}l^4 = \frac{32l^4}{\pi^6}\left[\frac{1}{1^6}+\frac{1}{3^6}+\frac{1}{5^6}+\cdots\right]$$

$$\therefore \frac{\pi^6}{960} = \frac{1}{1^6}+\frac{1}{3^6}+\frac{1}{5^6}+\cdots$$

(iii) Let $S = \frac{1}{1^6}+\frac{1}{2^6}+\frac{1}{3^6}+\frac{1}{4^6}+\cdots$

$$= \left(\frac{1}{1^6}+\frac{1}{3^6}+\frac{1}{5^6}+\cdots\right)+\left(\frac{1}{2^6}+\frac{1}{4^6}+\frac{1}{6^6}+\cdots\right)$$

$$\therefore S = \frac{\pi^6}{960}+\frac{1}{2^6}\left(\frac{1}{1^6}+\frac{1}{2^6}+\frac{1}{3^6}+\frac{1}{4^6}+\cdots\right) = \frac{\pi^6}{960}+\frac{1}{64}S$$

$$\therefore S - \frac{S}{64} = \frac{\pi^6}{960} \qquad \therefore \frac{63}{64}S = \frac{\pi^6}{960}$$

$$\therefore S = \frac{\pi^6}{960}\cdot\frac{64}{63} = \frac{\pi^6}{945}$$

$$\therefore \frac{\pi^6}{945} = \frac{1}{1^6}+\frac{1}{2^6}+\frac{1}{3^6}+\cdots$$

10. Obtain a half – range cosine series for $f(x)$, where $f(x) = \begin{cases} x & 0 < x < \pi/2 \\ \pi - x & \pi/2 < x < \pi \end{cases}$

Solution: Let $f(x) = \frac{a_0}{2} + \sum a_n \cos nx \qquad [\because\ l = \pi]$

$$\therefore a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \frac{2}{\pi}\left[\int_0^{\pi/2} x\,dx + \int_{\pi/2}^{\pi}(\pi - x)\,dx\right] = \frac{2}{\pi}\left[\left\{\frac{x^2}{2}\right\}_0^{\pi/2} + \left\{\pi x - \frac{x^2}{2}\right\}_{\pi/2}^{\pi}\right] = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\left[\int_0^{\pi/2} x\cos nx\,dx + \int_{\pi/2}^{\pi}(\pi - x)\cos nx\,dx\right]$$

$$= \frac{2}{\pi}\left[\left\{x\cdot\left(\frac{\sin nx}{n}\right) - (1)\cdot\left(-\frac{\cos nx}{n^2}\right)\right\}_0^{\pi/2} + \left\{(\pi - x)\cdot\left(\frac{\sin nx}{n}\right) - (-1)\cdot\left(-\frac{\cos nx}{n^2}\right)\right\}_{\pi/2}^{\pi}\right]$$

$$= \frac{2}{\pi}\left[\left\{\frac{\pi}{2}\cdot\frac{1}{n}\cdot\sin\frac{n\pi}{2} + \frac{1}{n^2}\cos\frac{n\pi}{2} - \frac{1}{n^2}\right\} + \left\{-\frac{1}{n^2}\cos n\pi - \frac{\pi}{2}\cdot\frac{1}{n}\sin\frac{n\pi}{2} + \frac{1}{n^2}\cos\frac{n\pi}{2}\right\}\right]$$

$$= \frac{2}{\pi n^2}\left[2\cos\frac{n\pi}{2} - \cos n\pi - 1\right]$$

$$\therefore a_1 = 0, a_2 = \frac{2}{\pi\cdot 2^2}[2(-1) - (+1) - 1] = -\frac{8}{\pi\cdot 2^2}, a_3 = 0, a_4 = 0, a_5 = 0$$

$$a_6 = \frac{2}{\pi\cdot 6^2}[2(-1) - (+1) - 1] = -\frac{8}{\pi\cdot 6^2}, a_7 = 0, a_8 = 0, a_9 = 0$$

$$a_{10} = \frac{2}{\pi\cdot 10^2}[2(-1) - (+1) - 1] = -\frac{8}{\pi\cdot 10^2}$$

$$\therefore f(x) = \frac{\pi}{4} - \frac{8}{\pi}\left[\frac{1}{2^2}\cos 2x + \frac{1}{6^2}\cos 6x + \frac{1}{10^2}\cos 10x + \cdots\right]$$

11. Obtain a half – range sine series in $(0, \pi)$ for $f(x) = x(\pi - x)$ and hence, find the value of $\sum \frac{(-1)^{n+1}}{(2n-1)^3}$

Solution: Let $f(x) = \sum b_n \sin nx$

$$\therefore b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x(\pi - x)\sin nx\,dx$$

$$= \frac{2}{\pi}\left[x(\pi - x)\left(-\frac{\cos nx}{n}\right) - (\pi - 2x)\left(-\frac{\sin nx}{n^2}\right) + (-2)\left(\frac{\cos nx}{n^3}\right)\right]_0^\pi$$
$$= \frac{2}{\pi}\left[\left\{0 - 0 - \frac{2}{n^3}\cos(n\pi)\right\} - \left\{0 - 0 - \frac{2}{n^3}\right\}\right]$$
$$= \frac{4}{\pi}\left[\frac{1-\cos n\pi}{n^3}\right] = \frac{4}{\pi n^3}[1 - (-1)^n]$$
$$= \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{8}{\pi n^3} & \text{if } n \text{ is odd} \end{cases}$$
$$\therefore x(\pi - x) = \frac{8}{\pi}\left[\frac{1}{1^3}\sin \pi x + \frac{1}{3^3}\sin 3\pi x + \frac{1}{5^3}\sin 5\pi x + \cdots\right]$$
Now, put $x = \pi/2$
$$\therefore \frac{\pi}{2}\left(\pi - \frac{\pi}{2}\right) = \frac{8}{\pi}\left[\frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots\right]$$
$$\therefore \frac{\pi^3}{32} = \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots \qquad \text{i.e. } \sum \frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi^3}{32}$$

12. Prove that in the interval $0 < x < \pi$, $\frac{e^{ax}-e^{-ax}}{e^{a\pi}-e^{-a\pi}} = \frac{2}{\pi}\left[\frac{\sin x}{a^2+1} - \frac{2\sin 2x}{a^2+2^2} + \frac{3\sin 3x}{a^2+3^2} - \cdots\right]$

Solution: Let $f(x) = e^{ax} - e^{-ax}$, $\quad 0 < x < \pi$, $\quad [\because l = \pi]$

Since, we want $f(x)$ to be expanded in sine terms only

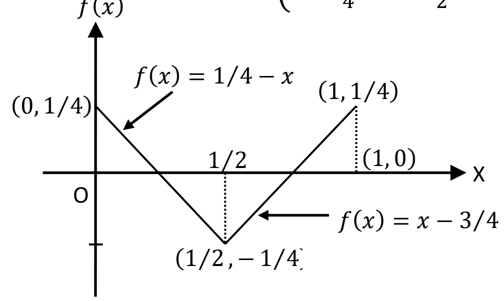
Let $f(x) = \sum b_n \sin nx$

$$\therefore b_n = \frac{2}{\pi}\int_0^\pi f(x)\sin nx\,dx = \frac{2}{\pi}\int_0^\pi (e^{ax} - e^{-ax})\sin nx\,dx$$
$$= \frac{2}{\pi}\left[\int_0^\pi e^{ax}\sin nx\,dx - \int_0^\pi e^{-ax}\sin nx\,dx\right]$$
$$= \frac{2}{\pi}\left[\left\{\frac{e^{ax}}{a^2+n^2}(a\sin nx - n\cos nx)\right\}_0^\pi - \left\{\frac{e^{-ax}}{a^2+n^2}(-a\sin nx - n\cos nx)\right\}_0^\pi\right]$$
$$= \frac{2}{\pi}\left[\left\{\frac{e^{a\pi}}{a^2+n^2}(-n\cos n\pi) + \frac{n}{a^2+n^2}\right\} - \left\{\frac{e^{-a\pi}}{a^2+n^2}(-n\cos n\pi) + \frac{n}{a^2+n^2}\right\}\right]$$
$$= \frac{2}{\pi}\cdot\frac{e^{a\pi}-e^{-a\pi}}{a^2+n^2}(-n\cos n\pi)$$

$$\therefore f(x) = \frac{2}{\pi}(e^{a\pi} - e^{-a\pi})\sum_{n=1}^\infty \frac{1}{a^2+n^2}(-n\cos n\pi)\sin nx$$
$$\therefore \frac{e^{ax}-e^{-ax}}{e^{a\pi}-e^{-a\pi}} = \frac{2}{\pi}\sum_{n=1}^\infty \frac{1}{a^2+n^2}(-n\cos n\pi)\sin nx$$
$$= \frac{2}{\pi}\left[\frac{1}{a^2+1}\sin x - \frac{2}{a^2+2^2}\sin 2x + \frac{3}{a^2+3^2}\sin 3x - \cdots\right]$$

13. Obtain the Fourier sine series expansion of the periodic function defined by $f(x) = \begin{cases} \frac{1}{4} - x, & if\ 0 < x < \frac{1}{2} \\ x - \frac{3}{4}, & if\ \frac{1}{2} < x < 1 \end{cases}$

Solution: The graph of the function $f(x)$:



Half-range Fourier sine series in the Half-range $(0, 1)$ is

$f(x) = \sum_{n=1}^\infty (b_n \sin nx)$ .... (A)

where $b_n = \frac{2}{\pi}\int_0^1 f(x)\cdot\sin\frac{n\pi x}{1}\,dx$

$$= 2\left\{\int_0^{1/2}\left(\frac{1}{4} - x\right)\cdot \sin n\pi x\, dx + \int_{1/2}^{1}\left(x - \frac{3}{4}\right)\cdot \sin n\pi x\, dx\right\}$$

$$= 2\left\{\left[\left(\frac{1}{4} - x\right)\cdot\left(\frac{-\cos n\pi x}{n\pi}\right) - (-1)\cdot\left(\frac{-\sin n\pi x}{(n\pi)^2}\right)\right]_0^{1/2} + \left[\left(x - \frac{3}{4}\right)\cdot\left(\frac{-\cos n\pi x}{n\pi}\right) - (1)\cdot\left(\frac{-\sin n\pi x}{(n\pi)^2}\right)\right]_{1/2}^{1}\right\}$$

$$= 2\left\{\left[\frac{1}{4n\pi}\left(\cos\frac{n\pi}{2}\right) - \left(\frac{\sin n\pi/2}{(n\pi)^2}\right) + \frac{1}{4n\pi}\right] + \left[-\frac{1}{4n\pi}(\cos n\pi) + \left(\frac{\sin n\pi}{(n\pi)^2}\right) - \frac{1}{4n\pi}\left(\cos\frac{n\pi}{2}\right) - \left(\frac{\sin n\pi/2}{(n\pi)^2}\right)\right]\right\}$$

$$= \frac{1}{2n\pi}[1 - (-1)^n] - \frac{4\sin n\pi/2}{(n\pi)^2}$$

To determine the value of $\sin\frac{n\pi}{2}$, three cases arise

Case 1: When n is even, $\sin\frac{n\pi}{2} = 0, (-1)^n = 1$ for which $b_n = 0$

Case 2: When n is odd, where $n = 1, 5, 9, \ldots$

$\sin\frac{n\pi}{2} = 1, (-1)^n = -1$ for which $b_n = \frac{1}{n\pi} - \frac{4}{\pi^2 n^2}$

Case 3: When n is odd, where $n = 3, 7, 11, \ldots$

$\sin\frac{n\pi}{2} = -1, (-1)^n = -1$ for which $b_n = \frac{1}{n\pi} + \frac{4}{\pi^2 n^2}$

Hence, (A) $\Rightarrow f(x) = \left(\frac{1}{\pi} - \frac{4}{\pi^2}\right)\cdot \sin \pi x + \left(\frac{1}{3\pi} + \frac{4}{3^2\pi^2}\right)\cdot \sin 3\pi x + \left(\frac{1}{5\pi} - \frac{4}{5^2\pi^2}\right)\cdot \sin 5\pi x + \cdots$

14. If a function $f(x)$ is defined in $(0, \pi)$ as $f(x) = \begin{cases} \frac{\pi}{3}, & 0 \le x \le \frac{\pi}{3} \\ 0, & \frac{\pi}{3} \le x \le \frac{2\pi}{3} \\ \frac{-\pi}{3}, & \frac{2\pi}{3} \le x \le \pi \end{cases}$

Find half – range **(a)** cosine series, **(b)** sine series of $f(x)$

Solution: (a) For cosine function, we have

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cdot \cos nx$

where $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\, dx = \frac{2}{\pi}\left\{\int_0^{\pi/3}\frac{\pi}{3}dx + \int_{\pi/3}^{2\pi/3} 0\, dx + \int_{2\pi/3}^{\pi}\frac{-\pi}{3}dx\right\}$

$= \frac{2}{\pi}\left\{\frac{\pi}{3}[x]_0^{\pi/3} - \frac{\pi}{3}[x]_{2\pi/3}^{\pi}\right\} = \frac{2}{\pi}\cdot\frac{\pi}{3}\left[\frac{\pi}{3} - \pi + \frac{2\pi}{3}\right] = 0$

$\therefore a_0 = 0$

$a_n = \frac{2}{\pi}\left\{\int_0^{\pi/3}\frac{\pi}{3}\cdot \cos nx\, dx + 0 + \int_{2\pi/3}^{\pi} -\frac{\pi}{3}\cos nx\, dx\right\}$

$= \frac{2}{\pi}\cdot\frac{\pi}{3}\left\{\left[\frac{\sin nx}{n}\right]_0^{\pi/3} - \left[\frac{\sin nx}{n}\right]_{2\pi/3}^{\pi}\right\} = \frac{2}{3n}\left\{\sin\frac{n\pi}{3} - \left[\sin n\pi - \sin\frac{2n\pi}{3}\right]\right\}$

$= \frac{2}{3n}\left[\sin\frac{n\pi}{3} + \sin\frac{2n\pi}{3}\right] = \frac{4}{3n}\sin\frac{n\pi}{2}\cdot\cos\frac{n\pi}{6}$

$\left\{\text{Using } \sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cdot\cos\left(\frac{C-D}{2}\right)\right\}$

When n is even $\sin\frac{n\pi}{2} = 0$ and therefore $a_n = 0$ for n even

We shall find first three non-zero terms for a_n: we have

$a_n = \frac{4}{3n}\sin\frac{n\pi}{2}\cdot\cos\frac{n\pi}{6}$

$n = 1, a_1 = \frac{4}{3}\sin\frac{\pi}{2}\cdot\cos\frac{\pi}{6} = \frac{4}{3}\frac{\sqrt{3}}{2} = \frac{2}{\sqrt{3}}$

$n = 3, a_3 = \frac{4}{3\cdot 3}\sin\frac{3\pi}{2}\cdot\cos\frac{\pi}{2} = 0$

$n = 5, a_5 = \frac{4}{3\cdot5}\cdot\sin\frac{5\pi}{2}\cdot\cos\frac{5\pi}{6} = \frac{4}{15}\left(-\frac{\sqrt{3}}{2}\right) = -\frac{2}{5\sqrt{3}}$

$n = 7, a_7 = \frac{4}{3\cdot7}\cdot\sin\frac{7\pi}{2}\cdot\cos\frac{7\pi}{6} = \frac{4}{3\cdot7}\frac{\sqrt{3}}{2} = \frac{2}{7\sqrt{3}}$

$n = 9, a_9 = \frac{4}{3\cdot9}\cdot\sin\frac{9\pi}{2}\cdot\cos\frac{3\pi}{2} = 0$ ....

Hence $f(x) = \frac{2}{\sqrt{3}}\left[\cos x - \frac{1}{5}\cos 5x + \frac{1}{7}\cos 7x - \cdots\right]$ is the required cosine series

(b) For sine function, we have

$f(x) = \sum_{n=1}^{\infty} b_n\cdot\sin nx$

where $b_n = \frac{2}{\pi}\left\{\int_0^{\pi/3}\frac{\pi}{3}\sin nx\,dx + 0 + \int_{2\pi/3}^{\pi} -\frac{\pi}{3}\cdot\sin nx\,dx\right\}$

$= \frac{2}{\pi}\cdot\frac{\pi}{3}\left\{\left[\frac{-\cos nx}{n}\right]_0^{\pi/3} - \left[\frac{-\cos nx}{n}\right]_{2\pi/3}^{\pi}\right\} = \frac{2}{3n}\left[-\cos\frac{n\pi}{3} + 1 + \cos n\pi - \cos\frac{2n\pi}{3}\right]$

$= \frac{2}{3n}\left[1 - \cos\frac{n\pi}{3} + \cos n\pi - \cos n\pi\cdot\cos\frac{n\pi}{3}\right]$

$\left\{\text{Since } \cos\frac{2n\pi}{3} = \cos\left(n\pi - \frac{n\pi}{3}\right) = \cos n\pi\cdot\cos\frac{n\pi}{3} + \sin n\pi\cdot\sin\frac{n\pi}{3} = \cos n\pi\cdot\cos\frac{n\pi}{3}\right\}$

$= \frac{2}{3n}\left(1 - \cos\frac{n\pi}{3}\right)(1 + \cos n\pi)$

So, when n is odd $(1 + \cos n\pi) = 0$ and therefore $b_n = 0$ for n odd

Now, we shall find first three non-zero terms for b_n. We have

$b_n = \frac{2}{3n}\left(1 - \cos\frac{n\pi}{3}\right)[1 + (-1)^n]$

$n = 2, b_2 = \frac{2}{3\cdot2}\left[1 - \left(\frac{-1}{2}\right)\right](2) = \frac{2}{6}\cdot\frac{3}{2}\cdot 2 = 1$

$n = 4, b_4 = \frac{2}{3\cdot4}\left[1 - \left(\frac{-1}{2}\right)\right](2) = \frac{2}{12}\cdot\frac{3}{2}\cdot 2 = \frac{1}{2}$

$n = 6, b_6 = \frac{2}{3\cdot6}[1 - 1](2) = 0$

$n = 8, b_8 = \frac{2}{3\cdot8}\left[1 + \frac{1}{2}\right](2) = \frac{2}{3\cdot8}\cdot\frac{3}{2}\cdot 2 = \frac{1}{4}$

Hence, the required sine series is $f(x) = \sin 2x + \frac{1}{2}\sin 4x + \frac{1}{4}\sin 8x + \cdots$

15. Obtain half – range sine series to represent $f(x) = \begin{cases} \frac{2x}{3}, & 0 \le x \le \frac{\pi}{3} \\ \frac{\pi - x}{3}, & \frac{\pi}{3} \le x \le \pi \end{cases}$

Solution: Half range Fourier sine series for $f(x)$ in $(0, \pi)$ is

$f(x) = \sum_{n=1}^{\infty} b_n\cdot\sin nx$

where $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cdot\sin nx\,dx$

$\therefore b_n = \frac{2}{\pi}\left[\int_0^{\pi/3}\frac{2x}{3}\cdot\sin nx\,dx + \int_{\pi/3}^{\pi}\left(\frac{\pi - x}{3}\right)\cdot\sin nx\,dx\right]$

$= \frac{2}{3\pi}\left\{\left[(2x)\left(\frac{-\cos nx}{n}\right) - 2\left(\frac{-\sin nx}{n^2}\right)\right]_0^{\pi/3} + \left[(\pi - x)\left(\frac{-\cos nx}{n}\right) - (-1)\left(\frac{-\sin nx}{n^2}\right)\right]_{\pi/3}^{\pi}\right\}$

$= \frac{2}{3\pi}\left\{\left[\left(\frac{2\pi}{3}\right)\left(-\frac{1}{n}\cos\frac{n\pi}{3}\right) + \frac{2}{n^2}\left(\sin\frac{n\pi}{3}\right) + \left[0 - \left(\frac{2\pi}{3}\right)\left(-\frac{1}{n}\cos\frac{n\pi}{3}\right) + \frac{1}{n^2}\left(\sin\frac{n\pi}{3}\right)\right]\right]\right\}$

$= \frac{2}{3\pi}\cdot\frac{3}{n^2}\cdot\sin\frac{n\pi}{3} = \frac{2}{\pi}\cdot\frac{1}{n^2}\sin\frac{n\pi}{3}$

Putting $n = 1, 2, 3, 4, 5, 6, \ldots$, we get,

$\therefore b_1 = \frac{2}{\pi}\sin\frac{\pi}{3} = \frac{2}{\pi}\cdot\frac{\sqrt{3}}{2}$

$b_2 = \frac{2}{\pi}\cdot\frac{1}{2^2}\sin\frac{2\pi}{3} = \frac{2}{\pi}\cdot\frac{1}{2^2}\cdot\frac{\sqrt{3}}{2}$

$b_3 = \frac{2}{\pi} \cdot \frac{1}{3^2} \sin \pi = 0$

$b_4 = \frac{2}{\pi} \cdot \frac{1}{4^2} \sin \frac{4\pi}{3} = \frac{2}{\pi} \cdot \frac{1}{4^2} \cdot \left(-\frac{\sqrt{3}}{2}\right)$

$b_5 = \frac{2}{\pi} \cdot \frac{1}{5^2} \sin \frac{5\pi}{3} = \frac{2}{\pi} \cdot \frac{1}{5^2} \cdot \left(-\frac{\sqrt{3}}{2}\right)$

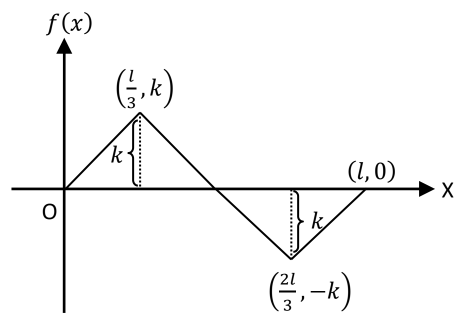
$b_6 = \frac{2}{\pi} \cdot \frac{1}{6^2} \sin 2\pi = 0$ and so on

Hence, the required Fourier series sine series for given $f(x)$ is

$$f(x) = \frac{2}{\pi} \cdot \frac{\sqrt{3}}{2} \left[\frac{1}{1^2} \sin x + \frac{1}{2^2} \sin 2x - \frac{1}{4^2} \sin 4x - \frac{1}{5^2} \sin 5x + \ldots\ldots\ldots\ldots\right]$$
$$= \frac{\sqrt{3}}{\pi} \left[\frac{1}{1^2} \sin x + \frac{1}{2^2} \sin 2x - \frac{1}{4^2} \sin 4x - \frac{1}{5^2} \sin 5x + \ldots\ldots\ldots\ldots\right]$$

16. Obtain Fourier sine series for half – range $(0, l)$ with period $2l$ for the function $f(x)$ represented

By $f(x) = \begin{cases} \frac{3kx}{l}, & 0 < x < \frac{l}{3} \\ \frac{3k}{l}(l - 2x), & \frac{l}{3} < x < \frac{2l}{3} \\ \frac{3k}{l}(x - l), & \frac{2l}{3} < x < l \end{cases}$



Solution: The Fourier sine series in the half-range $(0, l)$ is

$$f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{n\pi x}{l}\right)$$

where $b_n = \frac{2}{l}\int_0^l f(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx$

$$= \frac{2}{l}\left\{\int_0^{l/3} \frac{3kx}{l} \cdot \sin\left(\frac{n\pi x}{l}\right) dx + \int_{l/3}^{2l/3} \frac{3k}{l}(l - 2x) \sin\left(\frac{n\pi x}{l}\right) dx + \int_{2l/3}^{l} \frac{3k}{l}(x - l) \sin\left(\frac{n\pi x}{l}\right) dx\right\}$$

$$= \frac{6k}{l^2}\left\{\left[x\left\{-\cos\left(\frac{n\pi x}{l}\right)\frac{l}{n\pi}\right\} - \left\{-\sin\left(\frac{n\pi x}{l}\right)\right\}\frac{l^2}{n^2\pi^2}\right]_0^{l/3}\right.$$
$$+ \left[(l - 2x)\left\{-\cos\left(\frac{n\pi x}{l}\right)\frac{l}{n\pi} - (-2)\left\{-\sin\left(\frac{n\pi x}{l}\right)\right\}\frac{l^2}{n^2\pi^2}\right\}\right]_{l/3}^{2l/3}$$
$$\left.+ \left[(x - l)\left\{-\cos\left(\frac{n\pi x}{l}\right)\frac{l}{n\pi} - \left\{-\sin\left(\frac{n\pi x}{l}\right)\right\}\frac{l^2}{n^2\pi^2}\right\}\right]_{2l/3}^{l}\right\}$$

$$= \frac{6k}{l^2}\left\{\frac{-l^2}{3n\pi} \cdot \cos\left(\frac{n\pi}{3}\right) + \frac{l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right) + \frac{l^2}{3n\pi} \cdot \cos\left(\frac{2n\pi}{3}\right) - \frac{2l^2}{n^2\pi^2}\sin\left(\frac{2n\pi}{3}\right)\right.$$
$$\left.+ \frac{l^2}{3n\pi} \cdot \cos\left(\frac{n\pi}{3}\right) + \frac{2l^2}{n^2\pi^2}\sin\left(\frac{n\pi}{3}\right) - \frac{l^2}{3n\pi} \cdot \cos\left(\frac{2n\pi}{3}\right) - \frac{l^2}{n^2\pi^2}\sin\left(\frac{2n\pi}{3}\right)\right\}$$

$$= \frac{6k}{l^2} \cdot \frac{3l^2}{n^2\pi^2}\left[\sin\left(\frac{n\pi}{3}\right) - \sin\left(\frac{2n\pi}{3}\right)\right]$$

$$= \frac{18k}{n^2\pi^2} \cdot \sin\left(\frac{n\pi}{3}\right)[1 + (-1)^n]$$

.... $\left\{\text{Since } \sin\left(\frac{2n\pi}{3}\right) = \sin\left(n\pi - \frac{n\pi}{3}\right) = -(-1)^n \sin\left(\frac{n\pi}{3}\right)\right\}$

$\therefore b_n = 0,$ when n is odd

$= \frac{36k}{n^2\pi^2} \cdot \sin\left(\frac{n\pi}{3}\right),$ when n is even

$\therefore b_n = \frac{36k}{(2n)^2\pi^2} \sin\left(\frac{2n\pi}{3}\right)$ .... {Since n is even, it can be replaced by $2n$}

$= \frac{9k}{n^2\pi^2} \sin\left(\frac{2n\pi}{3}\right)$

Hence, the required half-range Fourier series is

$$f(x) = \sum_{n=1}^{\infty} \frac{9k}{n^2\pi^2} \sin\left(\frac{2n\pi}{3}\right) \cdot \sin\left(\frac{n\pi x}{l}\right)$$

The graph of the function $f(x)$ is

17. Find half range sine series for $f(x) = \frac{\pi}{4}$ in $(0, \pi)$. Hence show that

(i) $\frac{\pi}{4}\left(\frac{\pi}{2} - x\right) = \frac{1}{1^2}\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \cdots.$

(ii) $\frac{\pi}{8}x(\pi - x) = \frac{1}{1^3}\sin x + \frac{1}{3^3}sin3x + \frac{1}{5^3}\sin 5x + \cdots$

(iii) $\frac{\pi^3}{32} = \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots$

Solution: Half-range sine series for $f(x) = \frac{\pi}{4}$ in $(0, \pi)$ is

$f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin nx$

where $b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \cdot \sin nx \, dx$

$\therefore b_n = \frac{2}{\pi}\int_0^{\pi}\frac{\pi}{4} \cdot \sin nx \, dx = \frac{1}{2}\left[\frac{-\cos nx}{n}\right]_0^{\pi} = -\frac{1}{2n}[\cos n\pi - 1] = -\frac{1}{2n}[(-1)^n - 1]$

$\therefore b_n = \frac{1}{n}$ when n is odd

$= 0$ when n is even

Hence the required half-range series for $f(x) = \frac{\pi}{4}$ in $(0, \pi)$ is

$\frac{\pi}{4} = \sum_{n=1}^{\infty}\frac{\sin(2n-1)x}{(2n-1)}$

$\therefore \frac{\pi}{4} = \frac{1}{1}\sin x + \frac{1}{3}\sin 3x + \frac{1}{5}\sin 5x + \ldots\ldots\ldots\ldots$ .... (1)

Deduction:

(i) Integrate both sides of (1) w.r.t. x from 0 to x, we get,

$\frac{\pi}{4}\int_0^x 1\, dx = \frac{1}{1}\int_0^x \sin x\, dx + \frac{1}{3}\int_0^x \sin 3x\, dx + \frac{1}{5}\int_0^x \sin 5x\, dx + \ldots\ldots\ldots\ldots$

$\frac{\pi}{4}[x]_0^x = \frac{1}{1}[-\cos x]_0^x + \frac{1}{3}\left[\frac{-\cos 3x}{3}\right]_0^x + \frac{1}{5}\left[\frac{-\cos 5x}{5}\right]_0^x + \ldots\ldots\ldots$

$\therefore \frac{\pi}{4}x = \frac{1}{1^2}(1 - \cos x) + \frac{1}{3^2}(1 - \cos 3x) + \frac{1}{5^2}(1 - \cos 5x) + \ldots\ldots\ldots$ .... (2)

Now, put $x = \frac{\pi}{2}$ in (2), we get,

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots\ldots\ldots\ldots\ldots$ .... (3)

Subtracting (2) from (3), we get,

$\frac{\pi^2}{8} - \frac{\pi x}{4} = \frac{1}{1^2}\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \ldots\ldots\ldots\ldots\ldots$

$\frac{\pi}{4}\left(\frac{\pi}{2} - x\right) = \frac{1}{1^2}\cos x + \frac{1}{3^2}\cos 3x + \frac{1}{5^2}\cos 5x + \ldots\ldots\ldots\ldots\ldots$ as required .... (4)

(ii) Integrating (4) w.r.t. x from 0 to x, we get,

$\frac{\pi}{4}\int_0^x\left(\frac{\pi}{2} - x\right)dx = \frac{1}{1^2}\int_0^x \cos x\, dx + \frac{1}{3^2}\int_0^x \cos 3x\, dx + \frac{1}{5^2}\int_0^x \cos 5x\, dx + \ldots\ldots\ldots\ldots$

$\therefore \frac{\pi}{4}\left[\frac{\pi}{2}x - \frac{x^2}{2}\right]_0^x = \frac{1}{1^2}[\sin x]_0^x + \frac{1}{3^2}\left[\frac{\sin 3x}{3}\right]_0^x + \frac{1}{5^2}\left[\frac{\sin 5x}{5}\right]_0^x + \ldots\ldots\ldots$

$\therefore \frac{\pi x}{8}(\pi - x) = \frac{1}{1^3}\sin x + \frac{1}{3^3}\sin 3x + \frac{1}{5^3}\sin 5x + \ldots\ldots\ldots$ as required .... (5)

(iii) Put $x = \frac{\pi}{2}$ in (5), we get,

$\frac{\pi^3}{32} = \frac{1}{1^3} - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \ldots\ldots\ldots\ldots\ldots$ as required